Lab 4 – Report

Using and modifying the code provided at "lab3fixed", I tested the challenges that arise when Automatic Speech Recognition (ASR) systems encounter uncommon or fictional words. The confidence score served as an important indicator of how certain the recognizer was that the spoken utterance matched the transcribed text. The recognizer was evaluated using fictional names and places, scientific terminology and names of famous classical music composers.

I began by testing fictional places on the recognizer, using the relatively uncommon words "Rivendel" and "Tatooine." When I first said "Rivendel," the system's confidence score was approximately 0.12, meaning it was only about 12% certain that I had said "Rivendell." With each subsequent attempt, however, the recognizer became more confident. The score increased from 0.12 to 0.35, and finally reached 0.45 on my last attempt, suggesting that repeated exposure helped improve recognition. In contrast, the results for "Tatooine" were different. The system often misrecognized the word as "Tatouin" or "Tattooing," and the confidence level remained consistently low at around 0.21. Unlike "Rivendel," repeated attempts did not lead to any noticeable improvement in confidence.

Next, I tried fictional character names, specifically "Daenerys Targaryen" and "Voldemort." The first time I said "Daenerys," the system transcribed it correctly but with low confidence (around 0.26). Nevertheless, during the second and third attempts, it misrecognized the phrase as "The nearest Targaryen." Despite these transcription errors, the system's confidence gradually improved after multiple attempts, rising to 0.49 and eventually reaching 0.66. The opposite pattern appeared with "Voldemort." The system recognized the name accurately from the first attempt and showed relatively high confidence scores ranging from 0.62 to 0.75. This suggests that the recognizer was more familiar with this name.

Finally, I tested the botanical term "Lilium candidum," commonly known as the white lily, and the name of the Russian composer "Nikolai Rimsky-Korsakov." For "Lilium candidum," the system's confidence remained low despite repeated attempts, fluctuating only between 0.18 and 0.27, with no clear improvement. In contrast, "Nikolai Rimsky-Korsakov" began with a moderate confidence score of 0.38, but subsequent attempts produced inconsistent results, with scores either rising slightly to 0.40 or dropping to 0.31.

All these results raise questions about how effectively ASR can handle rare and unusual words. These systems face challenges when recognizing uncommon or fictional words because of how they are built. They rely on probability and data frequency, meaning that rare words, neologisms, and invented terms are less likely to appear in their training data and are therefore harder to recognize accurately. Additionally, a speaker's pronunciation and accent play an important role, as differences in acoustic patterns may not match what the model expects. This issue could be improved by incorporating a custom vocabulary, fine-tuning the model for specific domains, or implementing a confidence threshold that prompts the system to ask for clarification when the confidence score is low.